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RESEARCH ARTICLE

Multi-Translator FL: Modality-Agnostic Knowledge Construction via Modality Translation in Federated Learning

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ABSTRACT The paradigm of foundation models has shifted the technology trends through guiding a well-established model towards intelligence across a variety of modalities similar to human intelligence. At the same time, the growth of utilizing edge devices grows rapidly, acquiring multi-modal data at the client sides. However, despite this abundance of data and the unparalleled potential of development in artificial intelligence, the majority of current research has not taken into account effective multi-modal knowledge construction in the distributed resource-scarce device environments. This work aims to bridge the gap between the core ideas of multi-modal research (for yielding general knowledge from additional modalities) and the characteristics of distributed-settings (under limited devices with single modality). We propose a multi-translation federated learning architecture called MTFL, by making the server infuse modality-specific knowledge into a general modality-agnostic knowledge space via one-to-all translation as a bridge and reservoir, expanding the knowledge space across modalities. A single device is able to learn uncovered knowledge derived from a unified knowledge space at the server, while the server with a global model consisting of VAE encoder-decoders, performs modality-agnostic translation. Further, after constructing general modality-agnostic knowledge, MTFL translates the unified model back into modality-specific knowledge tailored to individual clients, facilitating the accommodation of various downstream tasks. The experimental results validate its effectiveness across multiple baselines using the CUB-200-2011, AwA2, Food-101, and MS-COCO datasets. Notably, on the CUB-200-2011, our MTFL performs 84.14% and 44.12% for image and text classification tasks, respectively; in the FL settings, with 82.42% and 42.77%.

INDEX TERMS On-device AI, federated learning, distributed learning, multi-modal learning, modality translation.

I. INTRODUCTION

The emergence of foundation models, such as large language models (LLM) can bridge existing knowledge based on visual modality, increasing capacity for comprehension across various modalities, like human intelligence. In addition, many research [2], [29], [43], [57] have showed that the models trained with several modalities could be more robust and intelligent rather than just single modality, for a variety of downstream tasks such as Visual Question Answering (VQA), cross-modal retrieval, visual speech generation.

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Additionally, co-learning is a concept under multi-modal research that is supported by resource-rich modality and represents resource-poor modality. This condition—such as insufficient or missing annotated data, incorrect labeling, or a lack of training or test modalities—occurs frequently in a real setting, even though it does not directly connect elements that are relevant to real-world scenarios, such as data privacy, distributed devices, and device-to-device communication. Concurrently, the growth of utilizing edge devices is expanding rapidly, resulting in a range of data from different domains, modalities, and device types. Real-world data produced by several sensors or devices might be used to tackle a variety of problems, including traffic, autonomous

driving, medical diagnostics, health care, and dynamic systems [1], [15]. However, even with this abundance of data and the unprecedented potential of AI models, it is not as easy as it may seem because multi-modal research works still struggle with a lack of multi-modal datasets, directly impeding progress in this area. One of the causes for this irony of having numerous data but not being able to use it, is the data from edge-devices are not being used properly, also known as data islands problem in the field of Federated Learning [52]. To put it another way, the majority of multi-modal research to date has not taken into account real-world situations, making it almost useless for the wealth of multi-modal data.

Federated Learning (FL) [33] is a clue to practically take advantage of isolated data from device in the distributed setting. According to FedAvg [33], Federated Learning in particular proposed using a wealth of data from mobile devices to train local models. Nonetheless, it is recognized that FedAvg [33] faces obstacles because of different data distributions, also referred to as the non-IID problem, which is one of the primary issues in FL research. The aforementioned issue is addressed in several previous works [20], [23], [41], [42], [47], but the heterogeneity brought about by various modalities is a recent addition to this field. Therefore, it is challenging for both multi-modal and FL research to address the variability induced by several modalities. Actually, multi-modal FL (MFL) is still in its early phases and is lagging behind in this field since multi-modal research is far ahead by employing a variety of techniques, including fusion, alignment, translation, and more. There have been a number of recent efforts at multi-modal FL [14], [27], [55], [56], including fusion mechanisms and contrastive learning. However, the majority of the attempts have the disadvantage of being predicated on unrealistic situations or susceptible to distinguishing between generalization and personalization, which we shall go into further depth about in the section on related works.

In this context, we start with the features for real-world environments where devices are frequently limited by resources [4], [39], and then develop a strategy to improve those subpar devices across several modalities. In other words, we need to get over edge devices' restrictions that prohibit them from utilizing multi-modality, which is a crucial idea; more modality would be ideal. Because of resource limitations, we begin by assuming that each device has a single modality. The difficulty is then examined in terms of how a device with a single modality might take multi-modality into consideration. Accordingly, we adopt the efficient approach of generating representation from the Variational Autoencoder [26], which aims to learn latent probability distribution and then to be used to construct the semantic feature, among the several methods for learning representation. Unlike the aforementioned methods, VAE is famous for its capacity to generate or sample features, which can help the model separate the features in latent space. Additionally, its variant, β -VAE, is widely recognized for its

ability to generate features and uncover hidden components in data, which may be advantageous for generalization and explainability [25], [49]. Using the VAE is essential to comprehend the generative components throughout the training process, including model, data, and fusion processes, as FL's objective is to achieve a balance between generalization and personalization. After all, with the demand for emulating real-world scenarios, we take it into consideration by collecting devices that serve distinct modality (such as vision or language) and then implementing the concept of multi-modality across them.

The main hypothesis that we have is as follows: *Can a global model serve as a universal knowledge space and then translate the knowledge back into a form of understandable model space with a specific uni-modality at each client?* More thoroughly, this framework relies on three assumptions: 1) the existence of distributed devices with uni-modal data; 2) the global model's ability to combine their representations into general knowledge, and then convert it back into uni-modal space; and 3) the learning process where each uni-modal device learns richer semantic information from its translated representation received from the server which could be useful for each personalization. This mechanism is similar to the established area of de facto multi-modal translation [4] which has been extensively studied in research. The primary distinction that this work possesses, however, is on modality-agnostic translation at the representation level, which enables the adaptation of various downstream tasks more easily. Finally, we propose a multi-translation federated learning architecture called MTLF that constructs general knowledge via the aggregation of modality-specific information, followed by translation that adapts to individual clients using a single modality. To simulate actual circumstances, we let MTLF operate within FL settings. Once the unified knowledge is established within this environment, each translator generates additional semantic knowledge adapting to the individual, which liberates the overall network from the type of modality.

To the best of our knowledge, Multi-Translator FL (MTFL) is the first work to enhance the uni-modal domain via representation translation without relying on explicit multi-modal models. The main contributions can be summarized as three parts:

- We elaborate on effective modality convergence through the unification of representations from uni-modal clients.
- This work provides a modality-agnostic balance from domain shift and modality gap through cross-modal knowledge translation.
- Our framework ultimately offers a scalable modality plug-in structure that preserves efficiency and generality across different modalities and architectures, as demonstrated through image-text validation.

We discuss related work on multi-modal FL and knowledge distillation in Sec. II, and propose our Multi-Translator FL framework (MTFL) in Sec. III. After validating our approach

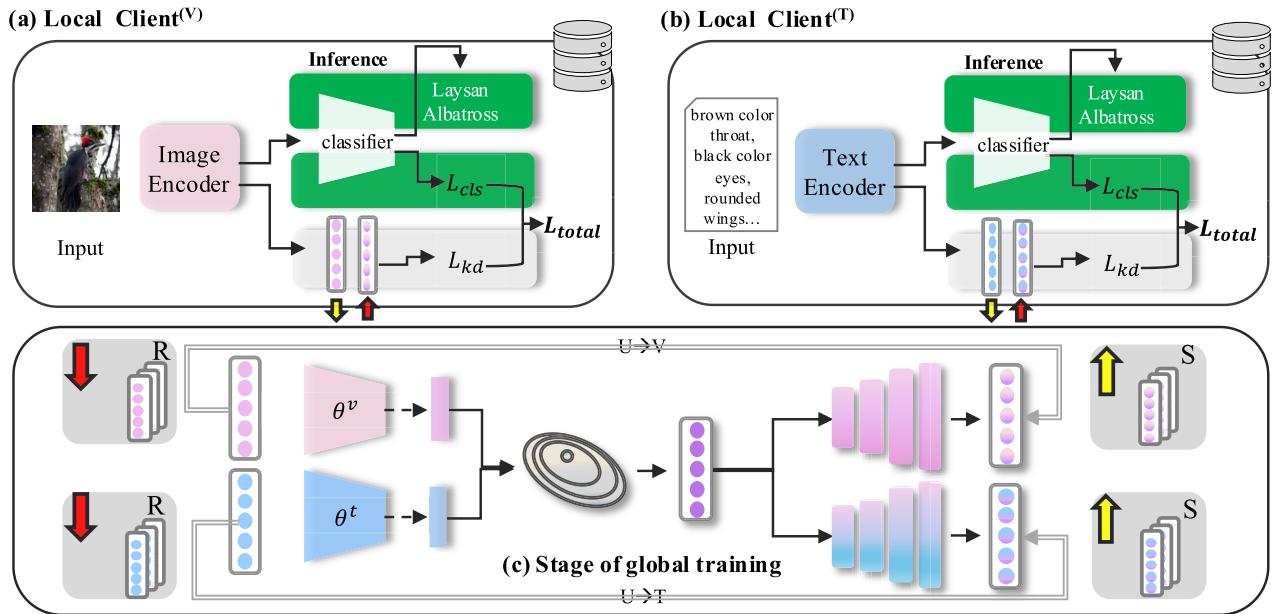


FIGURE 1. Two clients represented by the modality symbols V (image) and T (text) in the MTLF system framework. The server uses its encoder to unify the modality-specific information to the Universal (U) after receiving a feature from the client. It then employs a decoder to convert the universal knowledge back into a translated representation. For example, $U \rightarrow V$ is the translated representation for *Local Client^V*. The translated knowledge is provided to the client so that each client can benefit from it during transfer learning on client side.

for several tasks using related public datasets CUB-200-2011 [46], MS-COCO [30], AwA2 [51], and Food-101 [6] in Sec. IV, and conclude this paper in Sec. V.

II. RELATED WORK

Since this work is closely related to a cross domain of multi-modal federated learning and knowledge distillation, we discuss relevant recent work with strengths and limitations.

A. MULTI-MODAL FEDERATED LEARNING

Federated learning (FL) [33] enables mobile and edge devices to collaboratively train models in distributed environments, while preserving data privacy by keeping data localized on each device rather than sending raw data to a central server [22], [33]. Numerous research demonstrates that even relatively simple aggregation strategies, such as iterative parameter averaging with limited local optimization, work well in real-world scenarios. However, while most FL research has concentrated on uni-modal data situations, it has difficulties with heterogeneous data, particularly multi-modal datasets. To address the aforementioned issues, multi-modal federated learning (MFL) is being pursued. For example, [56] suggested a fusion strategy that falls within the category of decision-level fusion and uses representation to efficiently combine multi-modal information. Concentrating on the level of feature representation [11] also investigated the global representation by sharing the local encoder; however, it was limited to a single modality. In other ways, CreamFL [55] aims to connect the disparate modalities by constructing the unified feature space so that features with positive

pairs are close together, while those with negative pairs are the opposite. Above all, it is quite clear to investigate how features could be combined or fused to solve the problem from many modalities. However, most of the previously stated techniques, such as FedMultimodal [14] and CreamFL [55], might not be able to distinguish between the features related to personalization and generalization. Since learning generalization while preserving personal information is becoming more and more important for FL, the problem's significance is enhanced in distributed contexts.

Furthermore, it has been assumed in recent multi-modal federated learning (MFL) attempts that each client can access its local data using various modalities (depending on modality congruity) or with a partial set of modalities reflecting partial incongruity [9], [13]. In the actual world, nevertheless, it is impractical to expect that every device can offer the same range of modalities.

Another line of research states that in order to fill modality-gaps (for example, by complementing missing modalities at the client side), the central server must host a global multi-modal model capable of handling all modalities [55]. There is still a significant problem with MFL [8], [36] with how to deal with modality heterogeneity, where clients can only access a subset of all possible modalities, especially in single-modality. One of the primary issues with current MFL research topics is on connecting each client with a specific modality, which cannot be resolved by depending solely on global multi-modal modeling capabilities in the FL environment or a detour methodology.

B. KNOWLEDGE DISTILLATION

1) KNOWLEDGE DISTILLATION ACROSS MODALITIES

In the era of foundation models, which have already been trained on enormous datasets, transfer learning has emerged as a crucial strategy for utilizing the extensive knowledge included in these big models. Knowledge distillation (KD) is a branch of transfer learning that utilizes a Teacher-Student framework to impart general knowledge from the larger model to a smaller one. This field has broadly developed, shifting its focus from transfer strategies (such as logit-based, feature-based, and similarity-based schemes) to their successful adaptation in various downstream tasks [32]. More precisely, cross-modal adaptation has been studied [34], by transferring knowledge from a teacher's single modality (like an image) to student networks that contain knowledge from another modality (like a text), resulting in improved performance on specific tasks (like classification). In this case, the classic loss function is designed by combination of task loss and distillation loss, which forces the student model to learn prediction and feature alignment simultaneously. This method has exhibited the effectiveness of cross-modal adaptation via knowledge distillation for a range of downstream tasks, including continuous sign language recognition (CSLR) [16] and Semantic Segmentation [35]. Further, several research [21], [48] suggested that it might even be possible for students and teachers to communicate knowledge in both directions across modalities. Although distribution drift brought on by modalities' differences makes it impossible to achieve universal feature space alignment, cross-modal adaptation is still a challenging research problem.

2) KNOWLEDGE DISTILLATION IN FL

The interaction between the client and server, which is accomplished through iterative communication (known as a round in classic FL), is one of the fundamental strategies in FL environments. If the client's local models are large, there could be a significant communication cost overhead. To address these concerns, several studies have examined the effectiveness of knowledge distillation in FL settings [24], [50], [54]. FedKD has successfully demonstrated the way of adjusting KD into FL settings [50]. For instance, each client has a student model in addition to a global model, and knowledge distillation is updated solely through the student model. Even with smaller local models, it has shown efficacy in knowledge transmission. However, since it depends on uniform architectural settings, it has disadvantages in that it requires all clients to have the same global and student models.

III. PROPOSED METHOD

A. PROBLEM DEFINITION

This work addresses the problem of knowledge construction by multiple clients with access to single modal data in the federated learning scenario. It is assumed that low-resource devices gather data with a certain modality and perform local

training on a lightweight model. Their components can be separated by feature extractor and other modules, as common in personalized federated learning (PFL). The standard PFL approach separates the model into a task-specific head and a feature extractor to make it private or customized, and vice versa. However, it is challenging to gain from the expansion of multi-modal research, even when it is shareable during FL training because feature extractors (also called encoders) have never interacted with other modalities. With only one view in sight that is limited by a single modality, we thus recognize that the server model must act as a bridge between each encoder in order to expand the sight. Upon creating a unified knowledge space by combining the knowledge from each modality, we hypothesize that a single device can learn previously undiscovered knowledge generated from it. When a device is being fine-tuned and aided by unified knowledge, it may learn task-specific goals and optimize its encoder at the same time to capture meaningful expression.

B. MULTI-TRANSLATOR-BASED FL ARCHITECTURE

We present our Multi-Translator Federated Learning (MTFL) framework. By converting universal knowledge into modality-specific knowledge, we suggest a lightweight and modality-agnostic method. Additionally, we show that even by unifying uni-modal representations, modality translation is feasible at the representation level. Interestingly, our methodology compensates for missing client modalities without the need of relatively large extra multi-modal models on the local or even global side. Rather, just pre-trained uni-modal feature extractors on the local side may be effectively sufficient. The overall server-side training process can be divided into three steps: 1) to aggregate uni-modal representations from each local client into a hierarchical global encoder; 2) to generate a unified representation via variational inference; and 3) to construct translated representation for each local client from the global model at the server side as shown in (Fig. 2). The server side implements the first and second phases, while the local side contributes to the last step. The entire process serving as a unit of training can be divided into two or three parts, which correspond to binding the section where it is occurring or combining the goal of the process, respectively. In the following sections, each phase is covered in details. The whole training process is illustrated in (Fig. 1).

Preliminaries. In accordance with standard federated learning settings, we consider K clients, with each equipped with a uni-modal feature extractor. Feature-level representations which are generated by the extractor are forwarded to the server. Let $j_k \in \{1, \dots, J\}$ represent the learned modality for client k . Subsequently, each local client k ($k \in \{1, \dots, K\}$) possesses access to data from its respective modality, resulting in the storage of $(x_i, y_i)_{i=1}^{N_k}$, where each x_i belongs to the modality j_k as a component of the sampled dataset $\mathbb{X}$. Formally, we denote the local dataset of client K as $\{(x_i, y_i)\}_{i=1}^{N_k}$. For a given client k with uni-modality j , y^i

represents the corresponding label, and $(x_i, y_i)_{i=1}^{N_k}$ represents a sample from the modality j_k that client k handles. Both locally and globally, the entire modality data set $\mathbb{X} = \cup_{j=1}^J$ is hidden because each client only handles a single modality and does not share raw data. Also, the server model, which consists of a VAE-based encoder-decoder [26], starts a global training step when it receives only the client's abstract representations, $E_k(x^j) = z^{(j_k)}$, and not the complete raw modality data set $\mathbb{X} = \cup_{j=1}^J$. Ultimately, our two primary goals are separated into server and local side components. The aim of global training is to create unified representations (encoder) that can subsequently be converted back (decoder) into each local modality. The local model is then expected to learn a translated representation across several iterative rounds once the server model is ready to produce more semantic translated expressions.

Uni-modal aggregation. We first encode client's abstract representations, $E_k(x^j) = z^{(j_k)}$, while they operate on a classification assignment for each modality (for example, one client has a pre-trained model for categorizing bird species via photos, while another has a model based on textual descriptions of bird features). These are acquired by usage of their feature extractor, E_k , which is placed before the task-specific elements (i.e., classifiers). After that, the objective function for generating unified representation from uni-modal is based on maximizing the Evidence Lower Bound (ELBO) which is a lower bound to the marginal log-likelihood of the observed data, where it is impossible to compute the exact conditions of latent variables [5]. We clarify that our server model is typically composed of encoder-decoder structure. We design the modality-to-modality translator with the decoder because it bridges one modality-specific knowledge into another based on the sequence of encoder and decoder, interpreting as *modality translation*, albeit at the representation level in our scenario.

1) GENERALIZED MULTI-MODAL ELBO

Let $\mathbf{X} = \{\mathbf{x}^{(j)}\}_{j=1}^J$ be the set of J modalities.

$$\begin{aligned} \log p_{\theta}(\mathbf{X}) &\geq \mathcal{L}_{\text{ELBO}}(\theta, \phi; \mathbf{X}) \\ &= \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{X})} \left[\sum_{j=1}^J \log p_{\theta}(\mathbf{x}^{(j)} | \mathbf{z}) \right] \\ &\quad - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{X}) \| p(\mathbf{z})) \end{aligned} \quad (1)$$

We convert (1) to the generation of the representation level $\mathbf{Z}$, not the raw data $\mathbf{X}$, involving the parameters of the encoder ϕ and decoder θ for the global VAE model. Let $j_k \in \{1, \dots, J\}$ represent the client k 's modality. Consequently, the equation based on Product-of-Experts(PoE) [44] is refined as follows:

$$\begin{aligned} \log p_{\theta}(\mathbf{Z}) &\geq \mathcal{L}_{\text{ELBO}}(\theta, \phi; \mathbf{Z}) \\ &= \mathbb{E}_{q_{\phi}(\tilde{\mathbf{z}}|\mathbf{Z})} \left[\sum_{j=1}^J \log p_{\theta}(E_k(\mathbf{x}^{(j)}) | \tilde{\mathbf{z}}) \right] \\ &\quad - D_{\text{KL}}(q_{\phi}(\tilde{\mathbf{z}}|\mathbf{Z}) \| p(\tilde{\mathbf{z}})) \end{aligned} \quad (2)$$

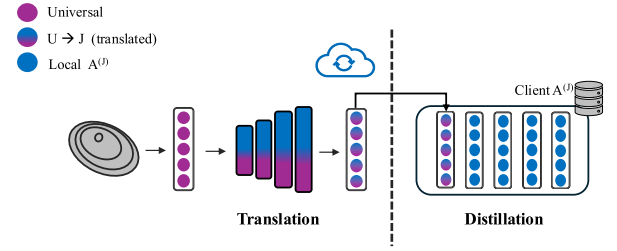


FIGURE 2. Target translation (global) and distillation (local).

2) HIERARCHICAL KL LOSS

The KL divergence loss between the fused posterior and the hierarchical prior is computed at each level l , ($l = 1, \dots, L$) in the server VAE model, which uses a hierarchical structure with L levels.

a: FUSED POSTERIOR

For J modalities $\{\mathbf{z}^{(j)}\}_{j=1}^J$, the approximate posterior in hierarchical layer levels is expressed with

$$\tilde{\mathbf{z}}_l | \{\mathbf{z}^{(j)}\}_{j=1}^J, \tilde{\mathbf{z}}_{l+1} \sim \mathcal{N}(\tilde{\mathbf{z}}_l; \mu_{q,l}, \text{diag}(\sigma_{q,l}^2)).$$

The parameters are computed via precision weighting:

$$\tau_{q,l} = \sum_{j=1}^J \tau_l^{(j)}, \quad \mu_{q,l} = \frac{\sum_{j=1}^J \mu_l^{(j)} \tau_l^{(j)}}{\tau_{q,l}}.$$

Here, the precision for each modality is defined as

$$\tau_l^{(j)} = \exp(-\log \sigma_l^{2(j)}), \quad j = 1, \dots, J.$$

3) TOTAL SERVER LOSS

Finally, the total loss of the server VAE is the sum of the modality-wise reconstruction losses between $\hat{\mathbf{z}}$ and $\mathbf{z}$, and the Fused Hierarchical KL divergence term computed from the aggregated representations (3). Essentially, it is followed by β -VAE [19], which uses a weighting hyper-parameter β to balance two elements of VAE (reconstruction and KL regularization).

$$\begin{aligned} \mathcal{L}_{\text{server}} &= \sum_{j=1}^J \underbrace{\text{MSE}(\mathbf{z}^{(j)}, \hat{\mathbf{z}}^{(j)})}_{\text{Translation Loss}} \\ &\quad + \beta \cdot \underbrace{\frac{1}{L} \sum_{l=1}^L D_{\text{KL}}(q_{\phi}(\tilde{\mathbf{z}}_l | \mathbf{Z}) \| p(\tilde{\mathbf{z}}_l | \tilde{\mathbf{z}}_{l+1}))}_{\text{Fused Hierarchical KL}} \end{aligned} \quad (3)$$

Modality translation. Afterwards, the server VAE, which is capable of generating more complex and semantically rich representations adapted to a specific single modality, is ready to deliver these representations to the clients. Each client follows the general framework of feature-based knowledge distillation [32], combining a distillation loss (e.g., mean squared error) with the task-specific loss (e.g., cross-entropy

loss). For a randomly selected client k , the local loss on each client is defined as follows:

$$\text{MSE}(Z_{\text{tr}}, E_k(x)) = \frac{1}{n} \sum_{i=1}^n (Z_{\text{tr},i} - E_k(x_i))^2 \quad (4)$$

$$\mathcal{H}(\text{MLP}(E_k(x))) = - \sum_{c=1}^C p(y_c | f) \log(p(y_c | f)) \quad (5)$$

Here, $\mathcal{H}$ denotes the softmax loss (6).

$$\mathcal{H}(y, f) = - \log(p(y|f)) = - \log\left(\frac{e^{f_y}}{\sum_{c=1}^C e^{f_c}}\right) \quad (6)$$

IV. EVALUATION

A. EXPERIMENT SETTING

1) DATASETS AND SETTINGS

We confirm our approach using six datasets listed below. Moreover, it is possible to separate those six datasets by merging the objective of our empirical research. First, we aim to confirm that (1) there is a performance improvement on both different modality clients, and (2) there is a performance improvement on non-iid conditions, finally, (3) we perform an ablation study to evaluate scalability and adjustment with various tasks. CUB-200-2011 [46], which contains 11,788 images of 200 bird species with detailed annotations for applications that require visual categorization with fine-grained information, is utilized for classification tasks for the purpose of (1). AWA2 [51] comprises 37,322 images in 50 animal categories, each labeled with 85 semantic attributes, and is utilized for the same purpose. We verify each vision Top-1 and text Top-1 accuracy for classification for both two datasets. In order to create text captions that explain the class label properties, we first collect the binary-based semantic features for both CUB-200-2011 and AWA2. In contrast to CUB, AWA2 attributes are determined by class labels instead of samples; hence, text augmentation is required during training under AWA2. Similarly, we utilized Food-101 [6], which comprises 101,000 real-world images organized into 101 food categories, for classification tasks. However, we apply a specific limitation when training models on the Food-101 dataset because it lacks language annotations compared to other datasets. To exploit text features, there is a common approach of Zero-shot settings for the Food-101 dataset, using a straightforward template (e.g., ‘A photo of a class’). Even more captivating is the fact that we see these Zero-shot settings resemble situations in FL, where some clients exhibit weak expression or where updates are not possible. Consequently, we assume that a client with inadequate textual expressions cannot update during Food-101 training, while a client with vision modality is permitted to do so. In more detail, the server establishes a unified latent space by using a rich image representation and a poor text representation with the common template, ‘A photo of a class.’ It subsequently transforms this into a vision-translated representation that will be utilized in transfer learning for the

Algorithm 1 Overall Training Process of MTFL

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1: INPUT: Client models  $\{E_k\}_{k=1}^K$ ; Server model  $\text{VAE}_{\phi, \theta}$ ;
   Number of server training epochs  $T_{\text{server}}$ ; Number of
   federated fine-tuning rounds  $T_{\text{local}}$ ; Number of federated
   rounds  $T_{\text{round}}$ ; Distillation weight  $\alpha$ 
2: OUTPUT: Trained client model parameters  $\{E_k\}_{k=1}^K$ 
3: for  $\text{round} = 1$  TO  $T_{\text{round}}$  do
4:   Stage 1: Server VAE Training
5:   for  $1$  TO  $T_{\text{server}}$  do
6:     Server randomly selects a subset of clients
      $S_t \subseteq \{1, \dots, K\}$ 
7:     // Start parallel execution
8:     for each client  $k \in S_t$  in parallel do
9:       Client freezes its model  $E_k$ 
10:      For a mini-batch of its local data  $\{(x_i, y_i)\}$ ,
        compute  $\mathbf{z}^{(jk)} \leftarrow E_k(x_i)$ 
11:      Send  $\mathbf{z}^{(jk)}$  to the server
12:    end for
13:    // End parallel execution
14:    Aggregate representations  $\{\mathbf{z}^{(jk)}\}_{k \in S_t}$  to form  $\mathbf{Z}$ 
15:    Compute server loss  $\mathcal{L}_{\text{server}}$ 
16:    Update VAE parameters  $\phi, \theta$ 
        by descending the gradient:  $\nabla_{\phi, \theta} \mathcal{L}_{\text{server}}$ 
17:  end for
18:  Stage 2: Client Model Fine-tuning
19:  Server freezes its VAE parameters  $\phi, \theta$ 
20:  Aggregate  $\{\mathbf{z}^{(jk)}\}_{k \in S_t}$  to form  $\mathbf{Z}$ 
21:  Generate unified latent representation  $\tilde{\mathbf{z}} \sim q_{\phi}(\tilde{\mathbf{z}}|\mathbf{Z})$ 
22:  for each modality  $j \in \{1, \dots, J\}$  do
23:    Generate translated representation  $\mathbf{Z}_{\text{tr}}^{(j)}$ 
24:  end for
25:  // Start parallel execution
26:  for each client  $k \in S_t$  in parallel do
27:    Server sends the translated representation  $\mathbf{Z}_{\text{tr}}^{(jk)}$  to
        client  $k$ 
28:    for  $1$  TO  $T_{\text{local}}$  do
29:      Get student representation  $\mathbf{z}_{\text{student}} \leftarrow E_k(x_i)$  for
        local data  $\{(x_i, y_i)\}$ 
30:      Compute  $\mathcal{L}_{\text{KD}} \leftarrow \text{MSE}(\mathbf{Z}_{\text{tr}}^{(jk)}, \mathbf{z}_{\text{student}})$  (Eq. 4)
31:      Compute  $\mathcal{L}_{\text{task}} \leftarrow \mathcal{H}(\text{MLP}(E_k(x_i)), y_i)$  (Eq. 5)
32:       $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \alpha \cdot \mathcal{L}_{\text{KD}}$ 
33:      Update  $E_k$  by descending the gradient:  $\nabla_{E_k} \mathcal{L}_{\text{total}}$ 
34:    end for
35:  end for
36:  // End parallel execution
37: end for

```

vision client. This scenario indirectly mimics the potential inconsistency of FL training, where some clients, referred to as stragglers in the FL [10], [40], [53], cannot participate because of communication problems, resource constraints, and other factors. Later on, we report the outcome of this scenario on evaluation with particular weak modality parts. We also utilize the CIFAR10/CIFAR100 datasets, which

comprise 32×32 color images with 6000 samples per class, with the exception of the number of classes (10 and 100). Using the CIFAR10/CIFAR100, (2) we assess how our methods could handle the common non-iid case. We want to demonstrate the adjustment on the case utilizing a common dataset that has been extensively studied by other FL research, even though our technique is intended to apply the concept of multi-modality across each distinct modality from device. Ultimately, we delve deeper into the paraphrasing task to confirm the objective of (3) using MS-COCO [30], which consists of 328,000 images and includes thorough object detection, segmentation, key-point detection, and captioning.

2) IMPLEMENTATION DETAILS

For the classification task on the CUB-200-2011, AwA2 datasets and Food-101, we employ BERT-Base Uncased [12] as the text backbone and ResNet-50 [18] as the vision backbone. In our experiment, we utilize each backbone as a feature extractor. Additionally, we use ViT-224 [37] and T5-Small [38] as the feature extractors to confirm the adaptation for the paraphrasing tasks. During model fine-tuning, the AdamW [31] optimizer is used with a learning rate of $1e-4$ for both FL and non-FL settings, while during global VAE training, $3e-4$ for FL and $2e-4$ for non-FL settings. In addition, we define a training round as the unit comprising five epochs for VAE training and five epochs for fine-tuning, except in FL-settings where fine-tuning is done over 3 epochs because of its complexity in FL. In more detail, we configured an IID setting with 10 clients and randomly select five clients for each communication round over a total of 20 rounds. For non-iid settings, we supply the simulation parameters using Table 3. We conduct all our experiments on a workstation that has an NVIDIA RTX 6000 Ada Generation GPU.

3) BASELINE METHODS

We evaluate using a number of baselines that fall into two categories: Non-FL and FL-settings. First, we quantify feature enhancement upon adopting our MTFL. We choose the Autoencoder based on concatenation and Cross-attention based fusion, which are considered essential algorithms in the field of Representation Learning for our Non-FL baselines. We compare MTFL with those three methods from Representation Learning on CUB-200-2011, AwA2 and Food-101 datasets.

For FL-settings, we compare our method with FedAvg [33] and transformed FedProx [41] under equal conditions, including formal requirements in FL environment (iid) like number of clients and number of participants per round. Please note that our baseline with FedAvg [33] shows the outcomes of applying algorithms for each modality (e.g., just vision, only text) corresponding to its intended purpose because those algorithms are based on homogeneous client settings.

In order to optimize transformed FedProx, which we refer to as P-FedProx, we decompose the architecture that

is later averaged for all of the clients without regard to modality and the others whose parameters are all averaged for modality-dependent. This method, which separates the model into sharable and non-sharable components during FL training, is predominant in the field of Personalized Federated Learning (PFL). To account for the modality-interaction and the weight average of the partial projection module from our diverse clients, we intend to apply the proximal term proposed in [41]. It enables us to introduce our clients to FedProx's fundamental idea. Lastly, by using FedAvg, which is represented as Fed-MTFL in Table 2, we simply extend our algorithm into FL settings. After that, our server averages the clients based on the modality type in addition to training the VAE models. Additionally, FedMultimodal [14], which is part of the multi-modal baselines for FL in the ablation research by Table 7, is included.

Exceptionally, we use different baselines during CIFAR10/CIFAR100 datasets for FL settings. Because the goal of the experiment with CIFAR10/CIFAR100 datasets is to confirm the capability of adaptation in general non-iid settings based on single modality. As previously proposed in pFedHN [42], we employ a number of baselines from Personalized Federated Learning (PFL).

B. CROSS-MODAL ADAPTATION

1) PERFORMANCE EVALUATION

As shown in Table 1, we evaluate our MTFL using CUB-200-2011 and AwA2 datasets and various benchmarks. With respect to classification accuracy, our approach surpasses the traditional representation strategy across all datasets and both modalities (vision and text). The cross-attention-based fusion method, which has been proven effective in fusion research, yields comparable results to our MTFL for both modalities in the CUB-200-2011 dataset, with slight differences of 0.33% and 0.25% for vision and text, respectively. In particular, our algorithm outperforms the performance of the Autoencoder strategy across all modalities of the CUB and AwA2 datasets. This outcome shows that utilizing VAE to create our semantic latent space is more thorough than merely using Autoencoder to reconstruct the features. Even though our MTFL shows the best results, the enhancement in performance for the AwA2 dataset is minor due to its already high baseline, akin to that of other baselines.

2) EVALUATION WITH SPECIFIC WEAK MODALITY

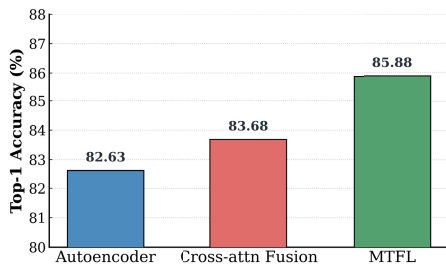
We investigate the effect of modality imbalance in which clients with vision modality updates, but with text modality under weak representation and model freezing. Experiments are executed on the Food-101 dataset, as shown in Figure 3. By achieving a classification accuracy of 85.88% on vision, which is higher than the 83.68% achieved through cross-attention fusion, we showcase the advantage. Moreover, compared to 82.63% from Autoencoder, the improvement gap is larger. It is interesting to see that our global VAE model successfully creates a unified latent space and is then

TABLE 1. Results of experiments for model components using CUB and AwA2 datasets. Both Text Top-1 and Vision Top-1 accuracy (%) are used to evaluate performance. The absolute performance gain over the baseline (Backbone-ft) is indicated by values in parenthesis.

Algorithm	Dataset			
	CUB-200-2011		AwA2	
	Vision top 1 (%)	Text top 1 (%)	Vision top 1 (%)	Text top 1 (%)
Backbone-ft (Baseline)	80.52	41.23	92.48	98.43
Autoencoder	83.56 (+3.04)	43.68 (+2.45)	93.08 (+0.60)	99.19 (+0.76)
Cross-attn Fusion	83.82 (+3.29)	43.87 (+2.64)	92.83 (+0.35)	99.26 (+0.83)
MTFL (Ours)	84.14 (+3.62)	44.12 (+2.89)	93.34 (+0.86)	99.31 (+0.88)

TABLE 2. Comprehensive comparison across federated learning algorithms on the CUB dataset. Performance (Vision/Text Top-1 Accuracy) is reported under the IID setting, along with communication overhead and computation cost per round.

Algorithm	Accuracy (%)		Communication Overhead (MB/round)			Computation Cost (sec/round)		
	Vision Top-1	Text Top-1	Upload	Download	Total	$T_l(s)$	$T_g(s)$	Total Times(s)
Global model(Initial)	80.52	41.23	—	—	—	—	—	—
FedAvg	81.87	42.60	2824.23	2824.23	5648.47	428.44	0.12	460.12
P-FedProx	81.97	42.58	2824.23	2824.23	5648.47	394.73	0.35	395.08
Fed-MTFL (Ours)	82.42	42.77	2824.23	2881.10	5705.33	464.43	158.40	622.83

**FIGURE 3.** Vision classification performance where clients with weak textual expression cannot update during Food-101 training, whereas a client utilizing the vision modality is allowed to perform an update. It aims to mirror the inconsistent aspects in real-world FL environments.

translated into specific modalities, even in a scenario where a client with weak text modality cannot perform an update. It is remarkable that our MTFL makes effective use of weak text modality, leading to improved performance for the local vision model that can be updated.

3) FL WITH IID SETTING

MTFL achieves the best classification results over baselines such as FedAvg and FedProx, with partial FedAvg indicated as P-FedProx on the CUB-200-2011 dataset, as shown in Table 2. MTFL shows improvements with the gap of 0.45 and 0.55 in vision classification accuracy for FedAvg and P-FedProx, respectively. Additionally, it presents the enhancement of 0.17, 0.19 compared to FedAvg and P-FedProx on text classification accuracy, indicating a slight

improvement. As previously stated in Sec. IV-A, we set all the elements for the IID setting to be equal: AdamW [31] with a learning rate of $1e-4$ is used, and 10 clients are randomly chosen with 0.5 fraction, and are trained locally over three epochs.

We also quantify the computation cost and communication overhead (MB/round) as trade-off to simulate FL environments in Table 2. Since every method we offer is based on the averaged parameters by FedAvg, the values for upload local models are equal. However, our clients also need to download translated information from VAE structure during local fine-tuning, which makes the download portion different from other algorithms. To be more specific, we use $Knowledge\ Size(BYTE) = N * FEATURE_DIM * FLOAT32_SIZE$ to determine knowledge size(BYTE). Here, N represents the number of total samples. Finally, we quantify the amount of knowledge at each round to be around 11.37MB for MTFL, which causes the overall communication cost to diverge from the FedAvg/p-FedProx. In other words, MTFL achieves better performance at a higher communication cost given the round by downloading the data for each client from the server. As in Figure 5, we further investigate the communication overhead by client count (5, 10, 20) in contrast to FedAvg/P-FedProx in FL simulation. Figure 5 illustrates how MTFL incurs a certain amount of transmission overhead while training with an increasing number of clients. It demonstrates our MTFL outperforms other FL approaches only with very marginal communication cost increase. We also show experimental time consumption results which are frequently utilized to

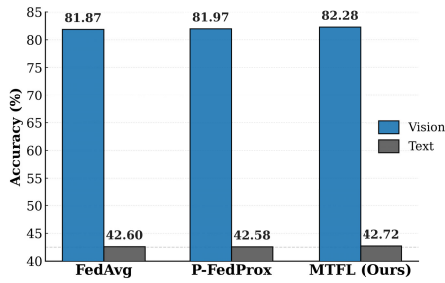


FIGURE 4. Comparison with other federated learning algorithms.

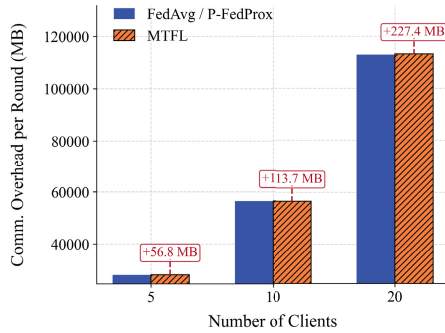


FIGURE 5. Computation overhead as the number of clients varies.

assess FL efficiency [7]. The local training and global training time consumptions are shown by T_l and T_g , respectively, in seconds.

The findings indicate that while the total time invested in local training is similar for all, the global training component differs. When compared to FedAvg, P-FedProx's approach necessitates more server time due to the global averaging being executed twice, for partial and global weights separately. Furthermore, our MTLF includes both VAE and averaging training, which requires extra computational efforts, making it the most resource-intensive on the server. Nevertheless, it is still manageable in comparison to using multi-modal models on a server due to their large number of parameters.

TABLE 3. Non-IID simulation parameters for CIFAR-10, CIFAR-100.

Sym.	Parameter	Value(s)
K	# of participating devices	(10, 50, 100)
N	# of nodes per client	2, 10 for CIFAR-10, 100
S	# of training steps	5000 steps for CIFAR-10, 100
IS	# of local inner steps	50 steps for CIFAR-10, 100
B	batch size	64
η_l	local learning rate	0.005
η_s	global(VAE) learning rate	0.0003

C. UNI-MODAL ADAPTATION

1) FL WITH NON-IID SETTING

Using the CIFAR10/CIFAR100 datasets, we demonstrate that MTLF can be used in the typical non-iid case.

We assign the two/ten classes to each client in accordance with pFedHN [42]. We then sample class-specific weights $a_{i,c} \sim \mathcal{U}(0.4, 0.6)$ for each client i in order to produce data heterogeneity. The classes are then assigned using the partitioning ratio $P_{i,c}$, which is defined as the normalized weight such that $\sum_c P_{i,c} = 1$. Further information on the whole non-iid architecture, including data partitioning, can be found in [42]. In a similar vein, we use the LeNet-based [28] local model with two convolutions from [42]. Everything else, with the exception of the server network, followed the environment described in pFedHN, which uses hypernetwork to execute the efficient personalization in FL, including a maximum of 5000 server-client communication steps, 50 local steps, and baselines for comparison, suggested in pFedHN are given in Table 3. The Federated Accuracy $\text{Acc} = \frac{1}{N} \sum_{i=1}^N \frac{1}{m_i} \sum_{j=1}^{m_i} \mathcal{L}_{\text{acc}}(f_i(x_j^{(i)}), y_j^{(i)})$, which is averaged across three seeds, is used to assess the models under the aforementioned circumstances. We use the conventional FL algorithm, such as FedAvg, as an acronym for the baselines; the remaining ones belong to the Personalized Federated Learning (PFL) category, as proposed in [42].

Lastly, Table 4 demonstrates that, even if MTLF is not designed for personalization, our approaches still produce a noteworthy outcome when compared to other baselines. Table 4 shows that our approach has the best accuracy with 10 clients in CIFAR-10. However, MTLF lags behind pFedHN in the remaining portion due to the latter's strong personalization capabilities against the non-iid settings. MTLF ranks the second overall, with the exception of the CIFAR-100 criterion with 100 clients. It was assumed that in order to satisfy the greater number of customer requirements, a more demanding design for VAE would be required.

D. ABLATION STUDY

1) EFFECT OF DIMENSION FOR LATENT SPACE IN VAE

As demonstrated in Table 5, we adjust the model configuration by altering the minimum dimension of the latent space in the VAE on the CUB dataset. We also investigate how the variable latent dimension interacts with the depth of VAE's text encoder in processing text features. This table shows the top results from six text encoder layers, each with 512 dimensions. Crucially, we set the vision encoder layers to four. Because we aim to analyze how performance varies with different VAE text encoder layer configurations and latent dimension sizes, while keeping the vision encoder condition constant. Given that the effect of increasing the number of vision encoder layers is not well understood, we will elaborate on it in Table 6.

Lastly, we discuss the effect of latent space dimension. MTLF needs additional layers to handle the text features received from the local client. Regarding the latent space dimension, it is vital to retain the dimension suitable to compress some level of meaningful expressions. In our case, when the latent dimension for VAE is fixed below 128, increasing the text layers for VAE does not result in a

TABLE 4. Effect of data heterogeneity on test accuracy over 10, 50, 100 clients on the CIFAR10 and CIFAR100 datasets.

Algorithm	CIFAR10			CIFAR100		
	10	50	100	10	50	100
Local	86.46 ± 4.02	68.11 ± 7.39	59.32 ± 5.59	58.98 ± 1.38	19.98 ± 1.41	15.12 ± 0.58
FedAvg	51.42 ± 2.41	47.79 ± 4.48	44.12 ± 3.10	15.96 ± 0.55	15.71 ± 0.35	14.59 ± 0.40
Per-FedAvg	76.65 ± 4.84	83.03 ± 0.25	80.19 ± 1.99	50.14 ± 1.06	45.89 ± 0.76	48.28 ± 0.70
FedPer	87.27 ± 1.39	83.39 ± 0.47	80.99 ± 0.71	55.76 ± 0.34	48.32 ± 1.46	42.08 ± 0.18
pFedMe	87.69 ± 1.93	86.09 ± 0.32	85.23 ± 0.58	51.97 ± 1.29	49.09 ± 1.10	45.57 ± 1.02
LG-FedAvg	89.11 ± 2.66	85.19 ± 0.58	81.49 ± 1.56	53.69 ± 1.42	53.16 ± 2.18	49.99 ± 3.13
pFedHN	90.83 ± 1.56	88.38 ± 0.29	87.97 ± 0.70	65.74 ± 1.80	59.48 ± 0.67	53.24 ± 0.31
MTFL(ours)	91.66 ± 1.05	87.29 ± 0.91	85.30 ± 0.46	63.88 ± 0.45	53.59 ± 0.84	46.05 ± 0.68

performance improvement. However, the greatest does not lead to improved performance either (this is represented by eight layers) due to some possible overfitting issue. This finding suggests that when planning to use latent space alongside various modalities, it is essential to consider a suitable depth of layers according to modality characteristics, while preserving sufficient space for information retention.

TABLE 5. Performance of local models after knowledge distillation training through MTFL with different VAE latent dimension and VAE text encoder layers on the CUB Dataset. Vision classification accuracy (V) and Text classification accuracy (T) results are shown across different numbers of text encoder layers and minimum latent dimensions. The best results for each metric are highlighted in bold.

Text Encoder Layers (#)	Metric	Minimum Latent Dimension			
		64	128	256	512
4	V (%)	83.09	83.75	83.88	83.98
	T (%)	43.52	43.77	44.07	43.27
6	V (%)	83.86	83.58	83.72	84.14
	T (%)	43.64	43.59	43.38	44.12
8	V (%)	83.75	83.70	84.62	84.00
	T (%)	43.64	43.80	43.61	43.48

TABLE 6. Performance of local models after knowledge distillation training through MTFL with different VAE vision encoder layers on the CUB Dataset. The best results (Top-1% acc) for each metric are highlighted in bold.

Vision Encoder Layers (#)	Vision(V) (%)	Text(T) (%)
4	84.14%	44.12%
6	84.27	43.41%
8	83.59%	43.34%

We further increase the number of VAE vision encoder layers to confirm the results, while maintaining the VAE text encoder layers fixed to 6 and the minimum VAE latent dimension maintained at 512. In contrast to the scenario of elevating the number of text encoder layers

TABLE 7. Performance on CUB multi-modal federated setting. Comparison across multi-modal or uni-modal federated learning algorithms on the CUB dataset. Performance (Top-1 Accuracy) is reported under the IID setting. All baselines are trained from scratch.

Configuration	Value
Dataset	CUB 200-2011
Number of clients	2
Total rounds	100
Local epochs	3
Local lr	2e-4
Local wd	5e-5
Global lr	3e-4
Batch size	64
Seed	42
Algorithm	Top-1 Accuracy (%)
FedAvg	80.74
FedMultimodal	80.75
mm-MTFL	81.91

from 4 to 6, Table 6 illustrates the effect of changing the number of VAE vision encoder layers, which appeared to be uncertain affection. Since it is commonly known that vision characteristics may have already included rich information, the effect of altering the number of VAE vision encoder layers may be negligible.

2) IMPACT OF MULTI-MODALITY

We examine the comparison with uni-modal and multi-modal FL in more details. We change our local client to a multi-modal client (mm-MTFL) for this configuration. Using our methods and FedMultimodal [14], which integrates the modality through a late-fusion, we adopt the baselines for multi-modal FL (MFL). Next, we choose FedAvg, a classic uni-modal based FL algorithm, for uni-modal. In a multi-modal network, we use ResNet-50 [18] / BERT-Base Uncased [12] to encode the image / text modality, whereas ResNet-50 [18] is only used for FedAvg (uni-modal). After that, we train all baseline networks from scratch using two randomly selected clients across three local epochs and a total of 100 rounds, adding five epochs for server VAE training specifically for mm-MTFL. Our approach obtains the best outcome, demonstrating the effectiveness of integrating the modality for VAE, as seen in Table 7. However, the advantage

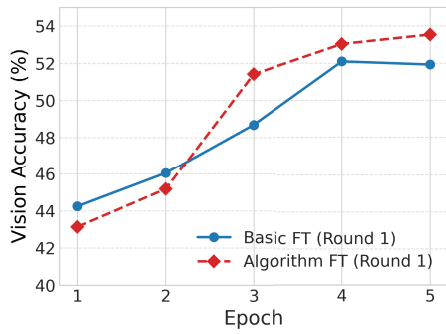


FIGURE 6. Performance comparison of vanilla fine-tuning on MS-COCO dataset.

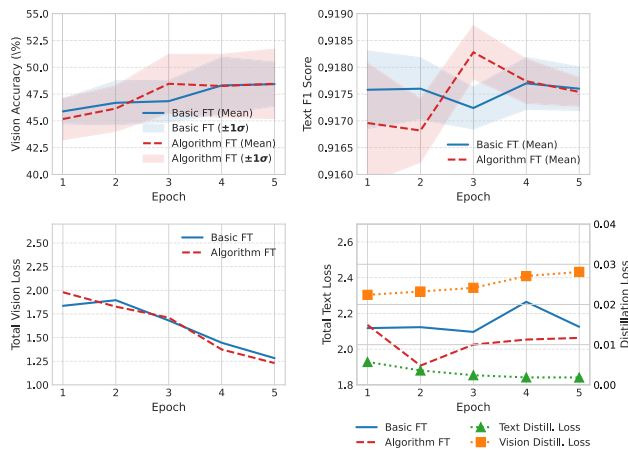


FIGURE 7. Epoch-wise average performance for five rounds. The top row shows the Vision accuracy (%) and Text F1 Score (%), while the bottom row shows how each portion of losses tends to alter in comparison to fine-tuned vanilla vision and text models.

of multi-modality for FedMultimodal over FedAvg is not evident. We speculate that the fusion technique is challenging to implement due to the complexity of the dataset, whereas our approach is less disrupted.

3) EXPLORING TO PARAPHRASING TASK

We examine MTLF in the context of evaluating the scalability of text paraphrasing beyond classification tasks. We intend to verify that the outcome of MTLF is better than that of each fine-tuned vanilla model, underpinned by the notion that performance can be enhanced through multi-modal training as opposed to uni-modal training. We choose the backbone architecture from vision transformer in this context to assess the robustness of various architectural designs. The subsequent step involves establishing Vanilla ViT (a non-multi-modal baseline) and Vanilla T5-small (also a non-multi-modal baseline) as the baselines for comparison. To compare with our algorithm, we fine-tune those models using a standard technique on the image-text pair subset of the MS-COCO [30] dataset. While text models paraphrase two given descriptions that are randomly selected from about five captions per image, vision models classify images into 80 categories based on the most often counted per sample.

Figure 6 displays results from the first round. Our MTLF achieves the best visual classification performance at 53.54%, surpassing the basic fine-tuned model, which has a performance of 52.09%. Additionally, as shown in the Figure 7, we perform the trend examination using Epoch-wise Averaging for a total of 5 rounds. We report the epoch-wise averaging performance $\pm 1\sigma$ on two plots in the top row to illustrate the text F1 score (%) and vision accuracy (%). It shows that our MTLF strategy begins with relatively higher initial total loss, but after epoch 3, MTLF becomes lower than the basic fine-tuned model. This outcome shows how the vision model has been enhanced by our method, enabling it to swiftly adapt to translated representation. It should be noted that the initial total loss is inevitably greater than that of the basic fine-tuned since we employ distillation loss to enable the model to learn translated representations. The model from our approach demonstrates how rapidly it adjusts to translated representation after the cross-over point, leading to a smaller loss composition. With the fine-tuned text model, the final plot also shows how our method rapidly adjusts to translated representation, lowering the text distillation loss over epochs 1 to 5. It is evident that the distillation process improves convergence stability and training efficiency.

E. DISCUSSION

1) BALANCE BETWEEN GENERALIZATION AND PERSONALIZATION

The MTLF's structure can possibly be separated into two components: latent space, which serves as general knowledge; and translator, which converts general information into specialized knowledge for each client. This implies that learning MTLF should lead to both generalization and personalization. However, there remain some interesting questions on how efficient and scalable it will be in terms of generalization and personalization, worthwhile to investigate as future works.

2) STRATEGY LEARNING VAE

In addition to the ablation study for investigating the number of layers for VAE, it is known that training the VAE is difficult [3], including the posterior collapse issue that frequently fails to disappear the KL divergence term. In this study, we use the approach from β -VAE and use an additional hyperparameter β ($\beta > 1$) to build a more factorized latent space. We search the hyperparameter for VAE including β , the number of encoding layers, learning rate, and others over the datasets. There is still room for improvement because some more recent advanced strategies for VAE, including using pre-trained VAE-based models [17], [45] can be applied.

V. CONCLUSION

This work explicitly connects the advantages of multi-modality with the limitations of distributed devices. Our proposed multi-translation federated learning approach,

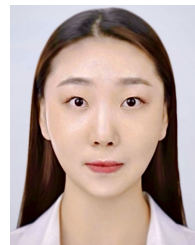
MTFL has demonstrated that a global model composed of VAE structures can be constructed to serve as a bridge across modalities. We have also proposed a translation framework based on the unified knowledge. This framework outlines general knowledge related to modality-agnostic translation at the representation level, which encompasses the functional modality plug-in tailored to specific clients. Our MTFL framework efficiently enables the local models with uni-modal to acquire more comprehensive knowledge from the server, based on limited resources. The framework allows us to improve performance on classification tasks and reduce training instability caused by modality differences. We have validated the model's scalability and evaluate its performance by investigating the paraphrasing task.

In light of this effort, we expect that our approach will assist with additional tasks and practical applications. Moreover, our work could assist in tackling particular problems like domain shift and cross-modal adaptation in FL. By integrating various modalities such as video, audio, and sensors, our research can be improved in the area of conceptual grounding. It can be focused on understanding the relationship between symbolic representation and perceptual function rather than depending only on performance metrics.

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